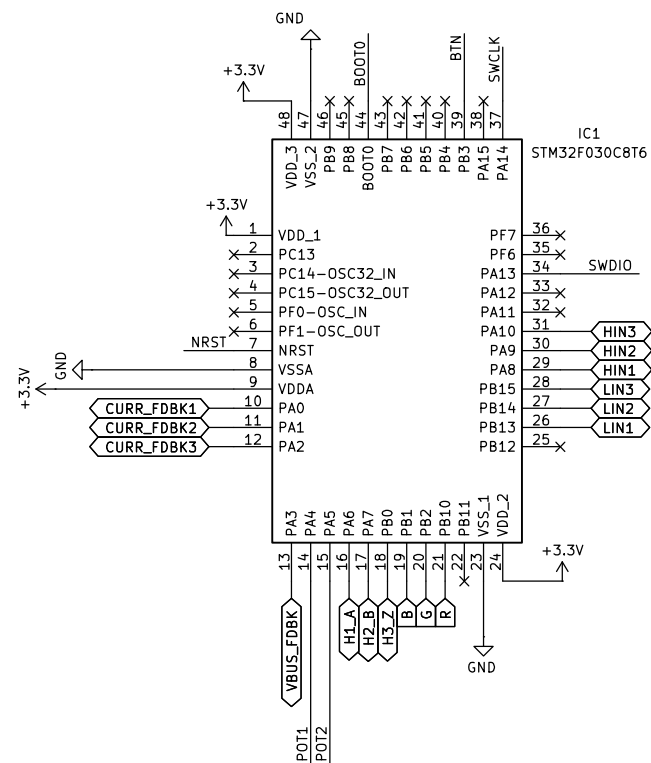
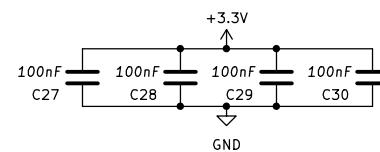
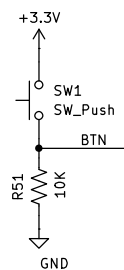
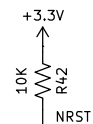
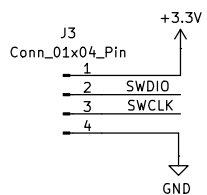
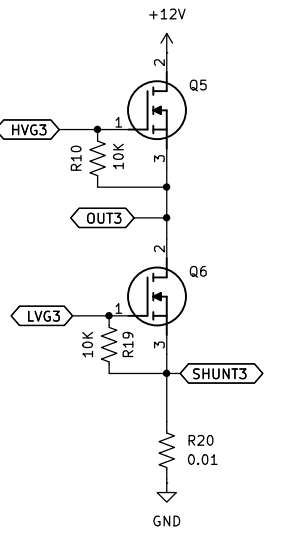
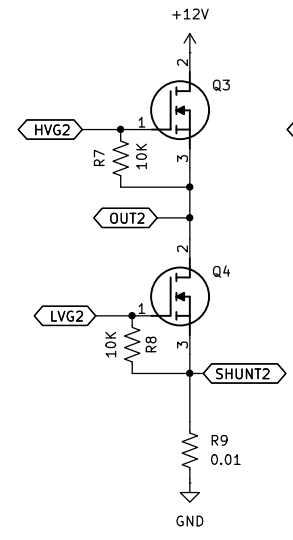
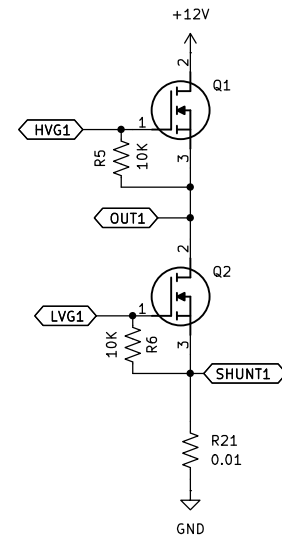
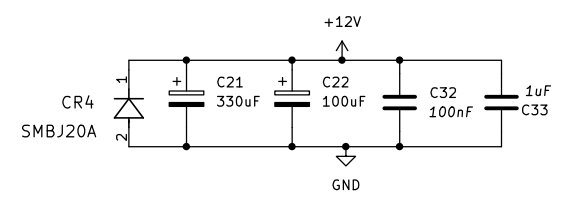
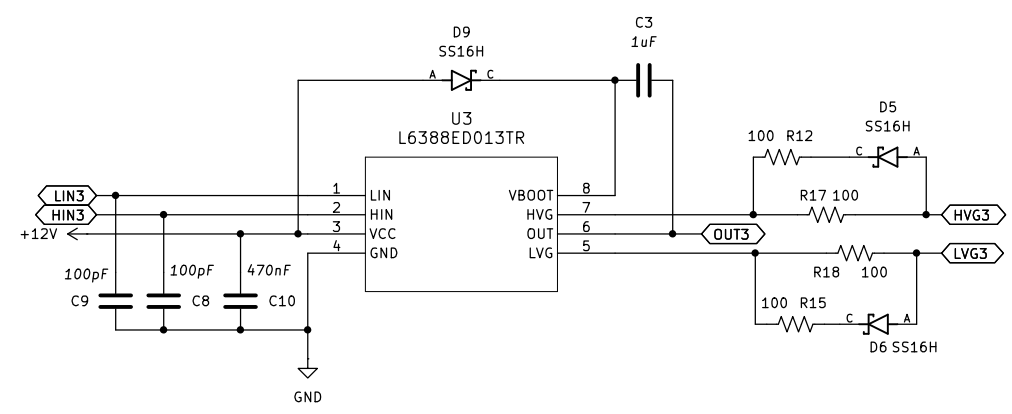
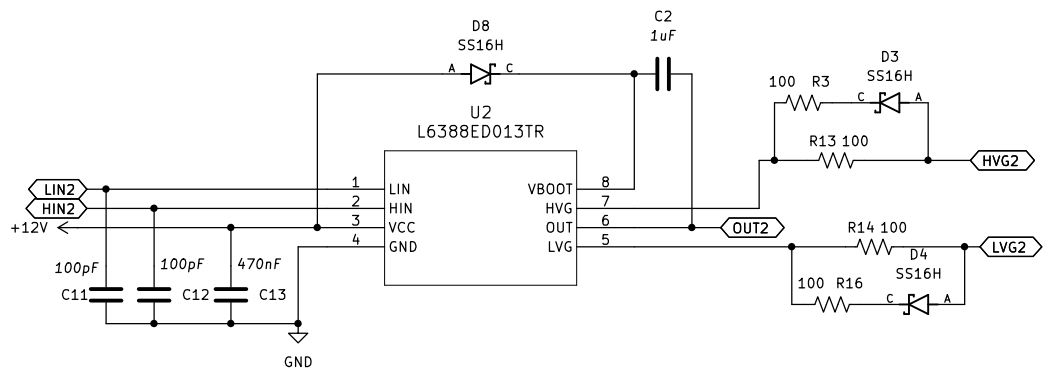
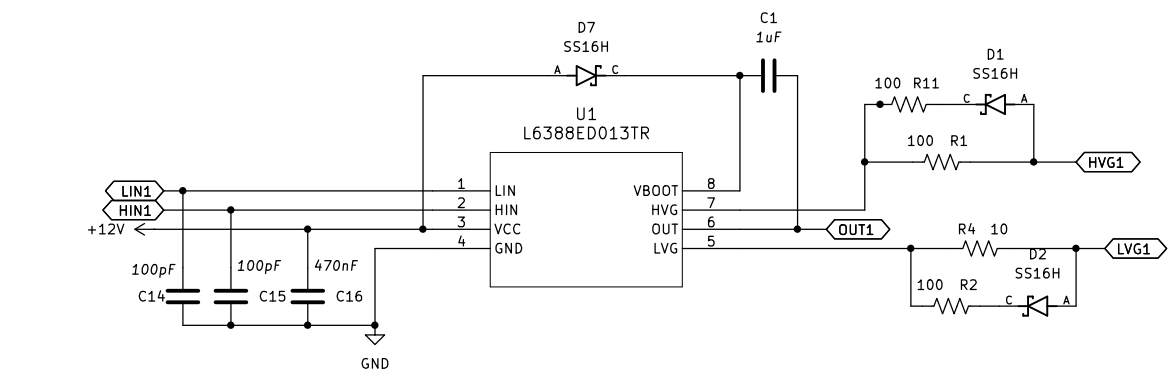
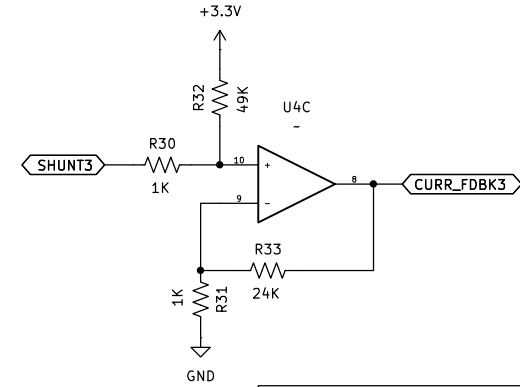
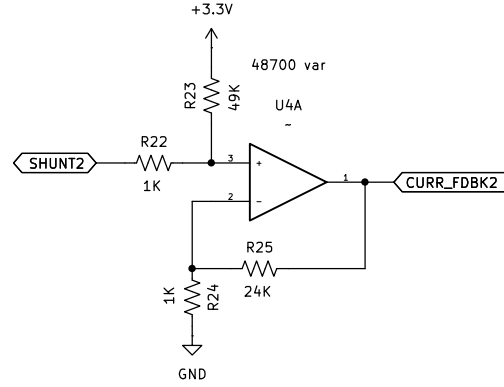
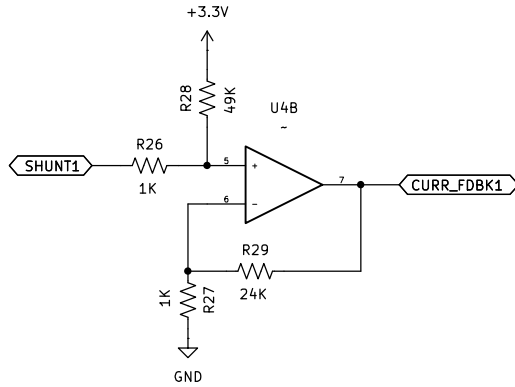
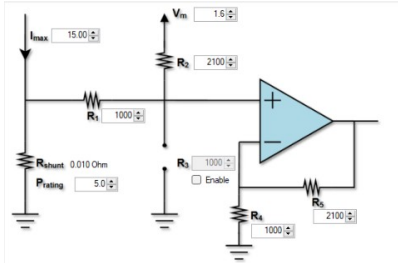
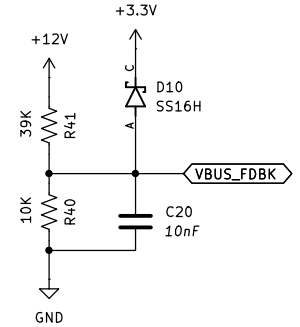
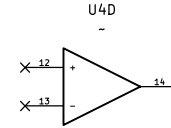
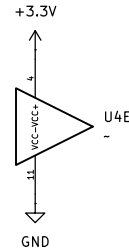








$$\text{Bus Voltage Divider} = (39K+10K)/10K = 4.9$$



$$\begin{aligned} \text{Gain} &= 1 + (2400/1000) = 25 \\ \text{Offset Network Attenuation} &= 48700/(48700+1000) = 0.980 \\ \text{Polarization Offset} &= 3.3 \cdot (1000/(1000+48700)) \cdot 25 = 1.66 \end{aligned}$$

Formüller
 $\text{Gain} = 1 + R_5 / R_4$
 $V_{\text{offset_input}} = V_{\text{ref}} \times R_1 / (R_1 + R_2)$
 $V_{\text{out_at_0A}}(\text{polarization offset}) = V_{\text{offset_input}} \times \text{Gain}$
 $\text{Offset Network Attenuation} = R_2 / (R_2 + R_1)$

Sayısal Hesaplar
 $0.01 \times 5A = 50\text{mV}$ (shunt üzerindeki max gerilim)
 $1.65 \pm 50\text{mV} \times 25(\text{gain}) = 2.9V - 400\text{mV}$ bu aralıkta değer üretilir
Polarization Offset = 1.65 ise R1 ve R2 kaçırır?
 $V_{\text{ref}} = 3.3V$
 $V_{\text{out_at_0A}} = 3.3V \times (1000 / (50000)) \times 25 = 1.65$
 $R_1 = 1K$
 $R_2 = 49K$

$$\text{Gain} = 1 + 24k/1k = 25$$

$$V_+ = V_{\text{shunt}} \times R_{28} / (R_{26} + R_{28}) + 3.3V \times R_{26} / (R_{26} + R_{28})$$

$$V_+ = V_{\text{shunt}} \times (49/50) + 3.3V \times (1/50)$$

$$V_+ = 0.98 \cdot V_{\text{shunt}} + 0.066V$$

$$0.066 \times 25 = 1.65, \text{ bu şekilde çıkıyor}$$

R1	1kΩ
R2	49kΩ
R4	1kΩ
R5	24kΩ
Rshunt	0.01

$$\begin{aligned} \text{Ratio} &= R_{\text{alt}} / (R_{\text{üst}} + R_{\text{alt}}) \\ V_{\text{Bus_sense}} &= V_{\text{in}} \times \text{Ratio} \\ \text{Ratio} &= 2.5V/12V \text{ (2.5V } \rightarrow \text{ADC için güvenli aralık)} \\ \text{Ratio} &= 0.208 \\ R_{\text{alt}} &= 10k \\ R_{\text{üst}} &= 38.08K \end{aligned}$$

$$\begin{aligned} R_{\text{üst}} &= 39k\Omega, R_{\text{alt}} = 10k\Omega \\ \text{Divider ratio} &= 10k / (39k + 10k) = 0.204 \\ 0.204 \times 12 &= 2.448 \\ \text{ADC kullanım: } &2.45V / 3.3V = 74\% \\ \text{Hassasiyet: } &3.95\text{mV/step} \\ \text{ADC max güvenli: } &3.7V \text{ (Schottky klempleme)} \end{aligned}$$

